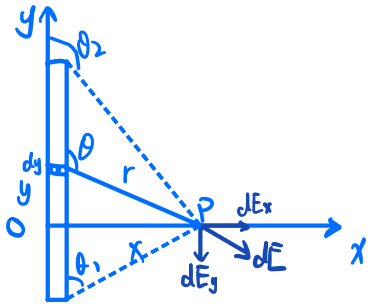
一、电场强度：

1. 长为L，带电量为q的均匀带电直杆，求P点电场强度.

解：



设线密度 $\lambda = \frac{q}{L}$

$dq = \lambda dy \quad dE = \frac{1}{4\pi\varepsilon_0}\frac{\lambda dy}{r^2}$

$dE_x = dE\sin\theta \quad dE_y = dE\cos\theta$

$$\begin{cases} r^2 = y^2 + x^2 = \frac{x^2}{\sin^2\theta} \\ y = -x\cot\theta \Rightarrow dy = x\sin^{-2}\theta\, d\theta \end{cases}$$

$dE_x = \frac{\lambda}{4\pi\varepsilon_0 x}\sin\theta\, d\theta \quad dE_y = \frac{\lambda}{4\pi\varepsilon_0 y}\cos\theta d\theta$

$E_x = \int_{\theta_1}^{\theta_2}\frac{\lambda}{4\pi\varepsilon_0 x}\sin\theta d\theta = -\frac{\lambda}{4\pi\varepsilon_0 x}(\cos\theta_2 - \cos\theta_1)$

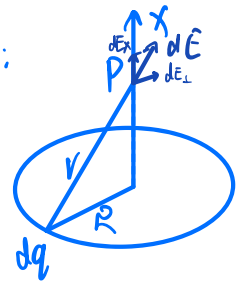
$E_y = \frac{\lambda}{4\pi\varepsilon_0 x}(\sin\theta_2 - \sin\theta_1)$

① $x \gg L$，杆可看成点电荷，$E_x = \frac{q}{4\pi\varepsilon_0 x^2} \quad E_y = 0$

② 无限长直导线 $\theta_1 = 0,\ \theta_2 = \pi$ $E_x = \frac{\lambda}{2\pi\varepsilon_0 x} \quad E_y = 0$

2. 半径为R均匀带电圆环，带电量为q，求轴线上P点电场强度

解：



$dq = \lambda dl \quad dE = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r^2}$

$dE_\perp = dE\sin\theta \quad dE_x = dE\cos\theta$

x轴对称 $\Rightarrow E_\perp = 0$

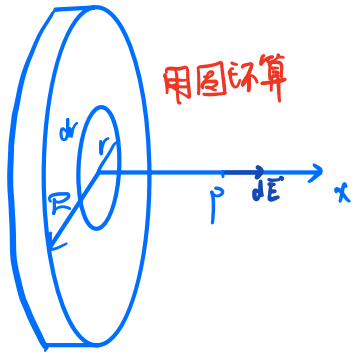
$E_x = \frac{1}{4\pi\varepsilon_0}\int\frac{dq}{r^2}\cos\theta = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\cos\theta$

$\cos\theta = \frac{x}{r} \quad r = (R^2 + x^2)^{\frac{1}{2}}$

$E = \frac{1}{4\pi\varepsilon_0}\frac{qx}{(R^2+x^2)^{\frac{3}{2}}}$

3. 面密度 σ，半径 R 的均匀带电圆盘在轴线上点 P 电场强度

解：



$dq = 2\pi r\,dr\,\sigma$

$dE = \frac{1}{4\pi\varepsilon_0}\frac{x\,dq}{(r^2+x^2)^{\frac{3}{2}}} = \frac{x\sigma}{2\varepsilon_0}\frac{r\,dr}{(r^2+x^2)^{\frac{3}{2}}}$

$E = \int dE = \frac{\sigma}{2\varepsilon_0}\left(1 - \frac{x}{(R^2+x^2)^{\frac{1}{2}}}\right)$

$\vec{E} = \frac{q}{2\pi\varepsilon_0 R^2}\left(1 - \frac{x}{(R^2+x^2)^{\frac{1}{2}}}\right)\vec{i}$

① $R \gg x$，无限大薄板，$E = \frac{\sigma}{2\varepsilon_0}$

② 有宽度 d 板，体密度 ρ，$E = \frac{\rho d}{2\varepsilon_0}$

4. 圆环电量 q，杆线密度 λ，长为 L，求杆对圆环作用力

解：

$dq = \lambda\,dx$

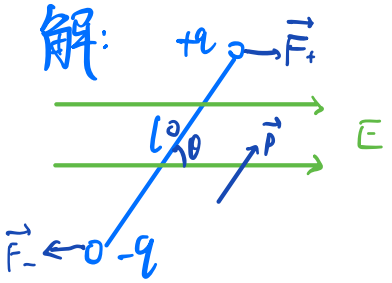
$E_x = \frac{1}{4\pi\varepsilon_0}\frac{qx}{(R^2+x^2)^{\frac{3}{2}}}$

$dF = E_x\,dq = E_x\,\lambda\,dx$

$F = \int_0^L \frac{qx\lambda\,dx}{4\pi\varepsilon_0(R^2+x^2)^{\frac{3}{2}}}$

5. 求电偶极子在均匀电场中受到的力偶矩。

解：



$\vec{F}_+ = q\vec{E}$，$\vec{F}_- = -q\vec{E}$

对 O：$M = F_+ \cdot \frac{1}{2}l\sin\theta + F_- \cdot \frac{1}{2}l\sin\theta = qlE\sin\theta$

$\vec{M} = q\vec{l}\times\vec{E} = \vec{p}\times\vec{E}$

① $\theta = \frac{\pi}{2}$，力偶矩最大　② $\theta = 0$，力偶矩为 0（稳定）

③ $\theta = \pi$，力偶矩为 0（不稳定）

二、高斯定理

1. 均匀电场中半径为R的半球面，求电通量

解：$\int_{半球}\vec{E}\cdot d\vec{S}+\int_{圆}\vec{E}\cdot d\vec{S}=0$

构成闭合面：$\int_{半球}\vec{E}\cdot d\vec{S}=-\int_{圆}\vec{E}\cdot d\vec{S}=\pi R^2E$

2. 均匀带电球面，电量Q，半径R，电场强度分布，

解：以球面为高斯面，电通量：$\oint\vec{E}\cdot d\vec{S}=\oint E\,dS=E\oint dS=E\cdot 4\pi r^2$

由高斯定理：$E\cdot 4\pi r^2=\frac{\sum q_{内}}{\varepsilon_0}$

在球外，$\sum q_{内}=Q$ $\therefore E=\frac{Q}{4\pi\varepsilon_0 r^2}$

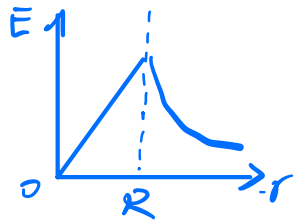
在球内，$\sum q_{内}=0$ $\therefore E=0$

3. 均匀带电球体，半径R，电荷体密度ρ，求电场强度分布.

解：以同心球面为高斯面：$\oint\vec{E}\cdot d\vec{S}=E\cdot 4\pi r^2=\frac{\sum q_{内}}{\varepsilon_0}$

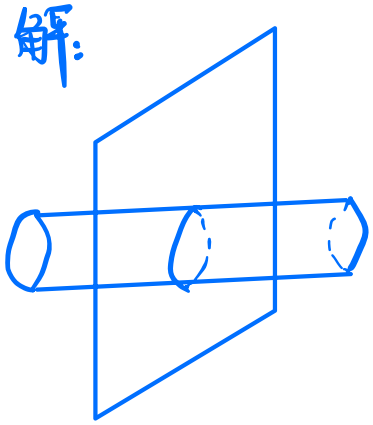
在球外：$(r>R)$ $\sum q_{内}=\frac{4}{3}\pi R^3\rho$ $\therefore E=\frac{\rho}{3\varepsilon_0}\frac{R^3}{r^2}$

在球内：$(r<R)$ $\sum q_{内}=\frac{4}{3}\pi r^3\rho$ $\therefore E=\frac{\rho}{3\varepsilon_0}r$



4. 无限大均匀带电平面，电荷面密度为σ，求电场强度分布

解：



选取垂直带电圆柱面为高斯面

$\Phi_e=\oint\vec{E}\cdot d\vec{S}=\int_{侧}\vec{E}\cdot d\vec{S}+\int_{左底}\vec{E}\cdot d\vec{S}+\int_{右底}\vec{E}\cdot d\vec{S}$

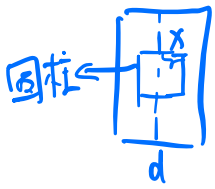
$=0+\int_{右底}E\,dS+\int_{右底}E\,dS$

$=2ES$

$\Phi_e=\frac{\sigma S}{\varepsilon_0}$ $\therefore E=\frac{\sigma}{2\varepsilon_0}$

5. 无限大均匀带电平板，电荷体密度 ρ，厚度 d，求电场强度分布

解：　选垂直且关于板对称的圆柱为高斯面

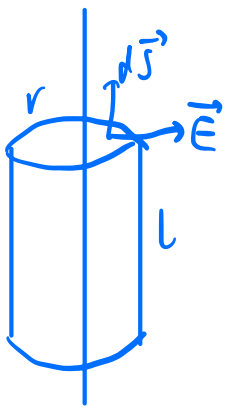


板外：$\Phi_e = 2ES = \frac{\rho S d}{\varepsilon_0}$　　$E_{外} = \frac{\rho d}{2\varepsilon_0}$

板内：$\Phi_e = 2ES = \frac{\rho S \cdot 2x}{\varepsilon_0}$　　$E_{内} = \frac{\rho x}{\varepsilon_0}$

6. 无限长 均匀带电直线，电荷线密度为 $+\lambda$，求电场强度分布

解：



$$\Phi_e = \oint \vec{E}\cdot d\vec{S}$$
$$= \int_{侧} \vec{E}\cdot d\vec{S} + \int_{上底} \vec{E}\cdot d\vec{S} + \int_{下底} \vec{E}\cdot d\vec{S}$$
$$= \int_{侧} E\cdot dS = E\cdot 2\pi r l$$

由高斯定理：$E\cdot 2\pi r l = \frac{\lambda l}{\varepsilon_0}$

$\therefore E = \frac{\lambda}{2\pi\varepsilon_0 r}$

三、电势、电势能：

1. 半径为 R，电量为 q 的均匀带电球体，求电势分布

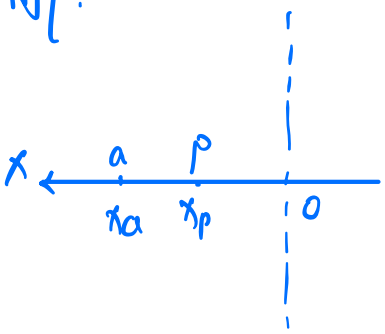
解：
$$\begin{cases} r < R & E_1 = \frac{qr}{4\pi\varepsilon_0 R^3} \\ r \geq R & E_2 = \frac{q}{4\pi\varepsilon_0 r^2} \end{cases}$$

对球外一点 P：$U_{外} = \int_P^\infty \vec{E_2}\cdot d\vec{r} = \int_r^\infty \frac{q\,dr}{4\pi\varepsilon_0 r^2} = \frac{q}{4\pi\varepsilon_0 r}$

对球内一点 P_1：$U_{内} = \int_{P_1}^\infty \vec{E_1}\cdot d\vec{r} = \int_r^R E_1 dr + \int_R^\infty E_2 dr = \frac{q}{8\pi\varepsilon_0 R^3}(3R^2 - r^2)$

2. 电荷线密度 λ 的无限长带电直线中电势分布。

解：

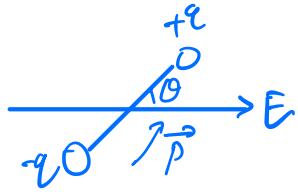


取 a 点为电势零点，

$$U_p = \int_{x_p}^{x_a} \frac{\lambda}{2\pi\varepsilon_0 x}\,dx$$

$$= \frac{\lambda}{2\pi\varepsilon_0}(\ln x_a - \ln x_p)$$

3. 求电偶极子在均匀电场中具有的电势能



解：

$$W = W_+ + W_- = qV_+ - qV_- = -q(V_- - V_+)$$

$$V_- - V_+ = \int_-^+ \vec{E}\cdot d\vec{l} = EL\cos\theta \Rightarrow W = -qEL\cos\theta$$

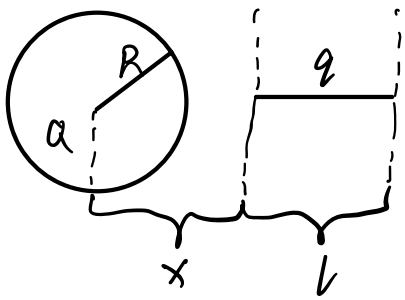
$$W = -\vec{p}\cdot\vec{E} \qquad \theta = 0 \Rightarrow W = -pE$$

4. 球半径 R，均匀带电量 Q，细杆长 l，均匀带电量 q

(1) 杆在带电球电场中具有的电势能

(2) 杆受到的电场力

(3) 杆左端运动到球面，电场力做功



解：(1) $V = \dfrac{Q}{4\pi\varepsilon_0 r}$

取 $dq = \lambda\,dr$ $\left(\lambda = \dfrac{q}{l}\right)$，无穷远为势能零点

$$dW = \frac{Q}{4\pi\varepsilon_0 r}\,dq = \frac{Q}{4\pi\varepsilon_0 r}\frac{q}{l}\,dr$$

$$\therefore W = \int_x^{x+l} \frac{Q}{4\pi\varepsilon_0 r}\frac{q}{l}\,dr = \frac{qQ}{4\pi\varepsilon_0 l}\ln\frac{x+l}{x}$$

(2) $$F = -\frac{\partial W}{\partial x} = \frac{qQ}{4\pi\varepsilon_0 l}\left(\frac{1}{x} - \frac{1}{x+l}\right) = \frac{qQ}{4\pi\varepsilon_0}\frac{1}{x(x+l)}$$

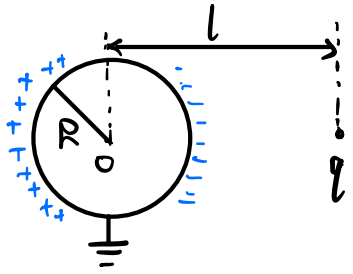
(3) $$W_1 = \frac{qQ}{4\pi\varepsilon_0 l}\ln\frac{x+l}{x}$$

$$W_2 = \frac{qQ}{4\pi\varepsilon_0 l}\ln\frac{R+l}{R}$$

$$A = W_1 - W_2$$

四、导体静电平衡：

1. 导体球附近有一点电荷q，求接地后导体上感应电荷的电量。

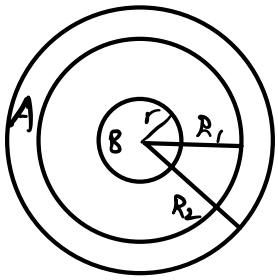


解：接地 $\rightarrow U=0$ → 等势体 O点电势也为0

$\therefore$ 在O点，$\frac{q}{4\pi\varepsilon_0 l}+\frac{Q}{4\pi\varepsilon_0 R}=0$ （Q → 感应电荷）

$\therefore Q=-\frac{R}{l}q$

2. 导体球壳A带电量Q，导体球B带电量q



(1) 将A接地再断开，电荷、电势分布

(2) 再将B接地，电荷、电势分布

解：(1) A接地时，内表面电荷 $-q$，全外表面电荷为 Q'

$U_A=\frac{1}{4\pi\varepsilon_0}\frac{Q'}{R_2}=0 \quad \therefore Q'=0$

A断开后，$U_A=0 \quad U_B=\frac{q}{4\pi\varepsilon_0 r}+\frac{-q}{4\pi\varepsilon_0 R_1}$

	(1)	(2)
r	q	q'
R_1	$-q$	$-q'$
R_2	0	$q'-q$

(2) 设B上电量 q'

$E_{内}=0 \Rightarrow Q_{内}=-q'$

$Q_{内}+Q_{外}=-q \Rightarrow Q_{外}=q'-q$

$U_B=\frac{q'}{4\pi\varepsilon_0 r}+\frac{-q'}{4\pi\varepsilon_0 R_1}+\frac{q'-q}{4\pi\varepsilon_0 R_2}$

$\therefore q'=\frac{qrR_1}{R_1r-R_2r+R_1R_2} \qquad U_A=\frac{q'-q}{4\pi\varepsilon_0 R_2}$

柱形电容器电场：$E=\frac{\lambda}{2\pi\varepsilon_0 r}$

平行板电容器：$w=\frac{W}{V}=\frac{1}{2}\varepsilon_0 E^2$

五、静电能：

1. 均匀带电导体球，带电量 Q，半径 R，求导体球的静电能.

解：$E=\frac{Q}{4\pi\varepsilon_0 r^2}$

$$W=\int_R^\infty \frac{1}{2}\varepsilon_0 E^2 dV$$
$$=\int_R^\infty \frac{1}{2}\varepsilon_0 \left(\frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\right)^2 4\pi r^2 dr$$
$$=\frac{Q^2}{8\pi\varepsilon_0 R}$$

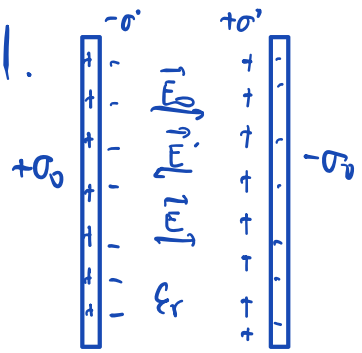
△带电体间相互作用能：

点电荷系：$W_{互}=\frac{1}{2}\sum_i q_i U_i$

电荷连续分布的静电能：$W=\frac{1}{2}\int_{(Q)} dq\,U$

六、静电场中的电介质：

1.



$$\sigma'=\left(1-\frac{1}{\varepsilon_r}\right)\sigma_0$$

适用条件：各向同性的均匀电介质充满整个空间，或电介质的表面为等势面

2. 电位移矢量：$\vec{D}=\varepsilon_0\varepsilon_r\vec{E}=\varepsilon\vec{E}$ （由正的自由电荷出发到负的）

电介质的高斯定理：$\oint_S \vec{D}\cdot d\vec{S}=\sum_i q_{0i}$

介质中的电场能量密度：$W=\frac{1}{2}EDV$ （普遍）

$w_e=\frac{1}{2}DE$ （有介质）

例1：两平行金属板之间充满相对介电常数为 ε_r 的各向同性均匀电介质，金属板上的自由电荷面密度为 σ_0，求两金属板之间的电场强度和介质表面的束缚电荷面密度。

解：$\int_S \vec{D}\,d\vec{S} = \sum_i q_{0i内} \Rightarrow D_0 S = \sigma_0 S$

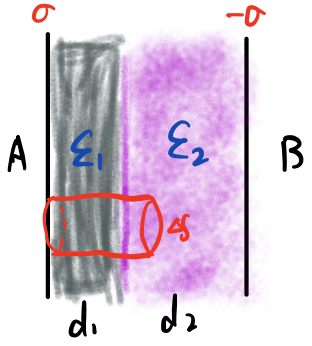
$\therefore D = \sigma_0$

$E = \frac{D}{\varepsilon} = \frac{\sigma_0}{\varepsilon_0 \varepsilon_r}$

$\int_S \vec{E}\,d\vec{S} = \frac{1}{\varepsilon_0}(\sigma_0 - \sigma')dS \Rightarrow E = \frac{1}{\varepsilon_0}(\sigma_0 - \sigma')$

$\therefore \sigma' = (1 - \frac{1}{\varepsilon_r})\sigma_0$

2. 平行板电容器，极板面积为 S，板间间距为 d，现充有两种电容率分别为 ε_1 和 ε_2，厚度分别为 d_1 和 d_2 的均匀电介质，如图。求此平行板电容器的电容。



解：取圆柱形高斯面，由高斯定理：

$D_0 S = \sigma_0 S \quad \therefore D = \sigma$

$\therefore E_1 = \frac{\sigma}{\varepsilon_1} \quad E_2 = \frac{\sigma}{\varepsilon_2}$

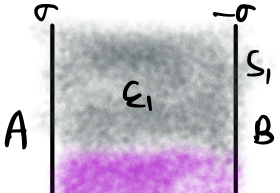
$\Delta V = \int_0^{d_1} \vec{E_1}\,d\vec{l} + \int_{d_1}^{d_1+d_2} \vec{E_2}\cdot d\vec{l}$

$= \frac{\sigma}{\varepsilon_0 \varepsilon_{r1}} d_1 + \frac{\sigma}{\varepsilon_0 \varepsilon_{r2}} d_2$

$C = \frac{q}{\Delta V} = \frac{\sigma S}{\sigma \frac{d_1}{\varepsilon_1} + \sigma \frac{d_2}{\varepsilon_2}} = \frac{\varepsilon_1 \varepsilon_2 S}{\varepsilon_1 d_2 + \varepsilon_2 d_1}$

↳ 两电容器串联

另一种情况：



$\sigma S = \sigma_1 S_1 + \sigma_2 S_2$

$D_1 = \sigma_1 \quad D_2 = \sigma_2$



$\Delta V_1 = \Delta V_2 = \Delta V$

$E = \frac{\Delta V}{d} \Rightarrow E_1 = \frac{\Delta V_1}{d} = E_2$

$D = \varepsilon E \Rightarrow \frac{\sigma_1}{\varepsilon_1} = \frac{\sigma_2}{\varepsilon_2}$

$\therefore \varepsilon_1 = \frac{\varepsilon_1 S}{\varepsilon_1 S_1 + \varepsilon_2 S_2}\sigma$ $\quad E_1 = E_2 = \frac{\sigma_1}{\varepsilon_1} = \frac{S}{\varepsilon_1 S_1 + \varepsilon_2 S_2}\sigma$

$\Delta V = E_1 d = \frac{Sd}{\varepsilon_1 S_1 + \varepsilon_2 S_2}\sigma$

$C = \frac{q}{\Delta V} = \frac{\varepsilon_1 S_1 + \varepsilon_2 S_2}{d}$ → 电容器并联

磁矩：$p = IS$　　磁力矩：$\vec{M} = \vec{p} \times \vec{B}$

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{1}{\varepsilon_0} \sum q$$

電場典型：

1. 直線：$E = \frac{\lambda}{2\pi\varepsilon_0 r}$

$E_y = \frac{\lambda}{4\pi\varepsilon_0 r}(\cos\theta_1 - \cos\theta_2)$

$E_x = \frac{\lambda}{4\pi\varepsilon_0 r}(\sin\theta_2 - \sin\theta_1)$

半無限長：$E_y = -\frac{\lambda}{4\pi\varepsilon_0 r}$，$E_x = \frac{\lambda}{4\pi\varepsilon_0 r}$

2. 球面：$E = \frac{Q}{4\pi\varepsilon_0 r^2}$ $(r \geq R)$　　$E = 0$ $(r < R)$

3. 球體：
   ① $\begin{cases} E = \frac{Q}{4\pi\varepsilon_0 r^2} & (r \geq R) \\ E = \frac{qr}{4\pi\varepsilon_0 R^3} & (r < R) \end{cases}$
   ② $\begin{cases} E = \frac{\rho}{3\varepsilon_0}\frac{R^3}{r^2} & (r \geq R) \\ E = \frac{\rho}{3\varepsilon_0} r & (r < R) \end{cases}$

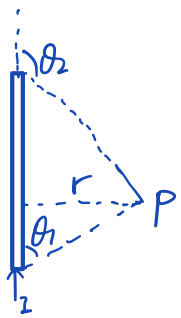
4. 平面：$E = \frac{\sigma}{2\varepsilon_0}$

5. 平板：$\begin{cases} E = \frac{\rho d}{2\varepsilon_0} & (\text{外}: x > \frac{d}{2}) \\ E = \frac{\rho x}{\varepsilon_0} & (\text{內}: x < \frac{d}{2}) \end{cases}$

$$\oint_L \vec{B} \cdot d\vec{l} = \mu_0 I$$

磁場典型：

1. 載流直導線：$B = \frac{\mu_0 I}{4\pi r}(\cos\theta_1 - \cos\theta_2)$



① 無限長：$B = \frac{\mu_0 I}{2\pi r}$

② 半無限長：$B = \frac{\mu_0 I}{4\pi r}$

2. 圓環：$B = \frac{\mu_0 I R^2}{2(R^2+x^2)^{\frac{3}{2}}}$　　圓心處：$B = \frac{\mu_0 I}{2R}$

3. 螺線管：$B = \mu_0 n I$

4. 螺線環：$B = \frac{\mu_0 N I}{2\pi r}$

5. 圓柱體：$\begin{cases} B = \frac{\mu_0 I}{2\pi r} & (r > R) \\ B = \frac{\mu_0 I r}{2\pi R^2} & (r < R) \end{cases}$

6. 平面：$B = \frac{1}{2}\mu_0 i$ ← 電流密度